

Do People Mirror Emotion Differently with a Human or TTS Voice? Comparing Listener Ratings and Word Embeddings

Michelle Cohn*
mdcohn@ucdavis.edu
University of California, Davis
Davis, California, USA

Grisha Bandodkar
gbandodkar@ucdavis.edu
University of California, Davis
Davis, California, USA

Raj Bharat Sangani
sanganir@uci.edu
University of California, Irvine
Irvine, California, USA

Kristin Predeck†
krispred@amazon.com
University of California, Davis
Davis, California, USA
Amazon AGI – Information
Santa Barbara, California, USA

Georgia Zellou
gzellou@ucdavis.edu
University of California, Davis
Davis, California, USA

ABSTRACT

Speakers increasingly communicate with technology using spoken language and display behaviors from human-human interaction, such as mirroring the pronunciation patterns of text-to-speech (TTS) voices. Yet, the magnitude and directionality of mirroring varies across prior studies comparing human and TTS addressees. The current study uses two approaches to assess mirroring of emotionally expressive speech produced by a human and a TTS voice. We compare AXB perceptual similarity (holistic human listener judgment, $n=109$ raters), and distance in latent space (wav2vec 2.0) on a dataset of single word shadowing in a cohort of 36 United States English speakers. Results show that both AXB and wav2vec 2.0 capture mirroring toward emotional prosody for both human and TTS voices. We discuss these findings in terms of theories of computer personification and emotional mirroring, as well as their contributions toward methodological advancements in phonetic entrainment.

CCS CONCEPTS

• **Human-centered computing** → **Empirical studies in HCI**; • **Applied computing** → **Psychology**.

KEYWORDS

Phonetic Mirroring, Emotional Expressiveness, Embeddings, Text-to-speech (TTS) Voices

ACM Reference Format:

Michelle Cohn, Grisha Bandodkar, Raj Bharat Sangani, Kristin Predeck, and Georgia Zellou. 2024. Do People Mirror Emotion Differently with a Human or TTS Voice? Comparing Listener Ratings and Word Embeddings. In *Extended Abstracts of the CHI Conference on Human Factors in Computing*

*Corresponding author

†This work was done prior to joining Amazon.

Systems (CHI EA '24), May 11–16, 2024, Honolulu, HI, USA. ACM, New York, NY, USA, 10 pages. <https://doi.org/10.1145/3613905.3650757>

1 INTRODUCTION

When people interact, they tend to adopt each other's behaviors, a process known as 'mirroring' (also referred to as 'alignment', 'entrainment', 'imitation', and 'accommodation') [26], including facial expressions [44] and even linguistic patterns, such as prosody [39]. A growing body of work has shown that people mirror linguistic patterns produced by technology, as well. For example, people adopt the words and syntactic structures produced by a computer system [9, 10] and the pronunciation patterns of text-to-speech (TTS) voices presented across a variety of forms [14, 17, 19, 24, 30, 49, 54, 56, 57]. However, the magnitude of mirroring often differs when making direct comparisons between a human and technological interlocutor. In speech, converging evidence suggests that people more strongly mirror other human voices than TTS voices [14, 31, 57]

In the current study, we test whether people mirror emotional expressiveness in human and TTS voices in distinct ways. Mirroring emotion, such as through facial expressions and speech, has been proposed as a mechanism to both experience and convey empathy [33]. While work in the field of affective computing has explored the impact of introducing emotional expressiveness by computers [11, 13, 34] and decoding people's emotional states and sentiments [7, 51, 52, 55], less research has examined how emotional expressiveness produced by human versus computer interlocutors shapes perception and behavior. The current study compares how people mirror emotion conveyed through the speech patterns of a human and a TTS voice, comparing two assessment methods. Overall, we find converging evidence for emotional alignment toward expressiveness to human and TTS voices by both human raters (Experiment 1) and in distance in latent space (Experiment 2).

1.1 Related Work

In the following sections, we provide an overview of theoretical accounts of emotional mirroring (Section 1.1.1), as well as an overview of methods previously used to assess acoustic similarity: individual acoustic features (Section 1.1.2), human perceptual similarity ratings (Section 1.1.3), and word embedding distances (Section 1.1.4).

1.1.1 Theoretical Accounts of Emotional Mirroring. Theoretical accounts of emotional mirroring, as a way to display empathy, differ in their specificity. On the one hand, some accounts propose that emotional mirroring is an unmediated, matched motor response [22, 23, 48] wherein a person sees and/or hears another person produce emotion, and they mirror it. For example, hearing smiled speech leads listeners to engage their zygomatic muscles, producing micro-smiles themselves [1]. Recent work provides some preliminary evidence for a matched motor response in emotional mirroring, including showing that speakers mirror emotionally expressive productions of words more than those words in a neutral tone produced by an Amazon’s Alexa TTS voice [21]. Broadly, these patterns are consistent with theories of technology equivalence [37, 42, 43], wherein people automatically and subconsciously apply patterns of behavior from human-human interaction to technology. On the other hand, some researchers propose that emotional mirroring is shaped by the social variables present between interlocutors [25, 32, 33]. For example, there is stronger mirroring toward in-group members than to out-group members [36], and weaker mirroring toward TTS voices might reflect these social groupings. Relatedly, converging evidence suggests greater (non-emotional) mirroring toward human voices than TTS voices [14, 31]. For example, perceptual sensitivity to artificially increased expressions of happiness differs for human and TTS voices [20, 29], suggesting that social categories might be an important factor in emotional mirroring in human-computer interaction.

1.1.2 Feature-based Approaches to Test Similarity. In assessing degree of speech mirroring, including for emotional prosody, groups often, a priori, pinpoint an acoustic-phonetic feature or set of features to examine. For example, mirroring has been observed for many features in human-human interaction, including speech rate and duration [18, 53] and pitch [19, 39]. In human-computer interaction, the degree to which the system mirrors the user is correlated with higher trust in chatbot interactions [38], greater engagement with a robot in a game context [50], and even better learning outcomes in tutoring paradigms [35, 40, 54].

Additionally, there are feature-based approaches used to examine the human user’s mirroring of a computer system. [18] found that participants engaging with a social chatbot (Amazon Alexa Prize) mirrored the system’s speech rate in a turn-by-turn fashion. Relatively less work has made a direct comparison of human mirroring toward computer versus human interlocutors. For example, [31] found that participants adopt the pronunciation of a tutoring system similarly when the system produced a pre-recorded human or TTS voice. In examining mirroring of emotional expressiveness, [19] had participants shadow interjections (e.g., “awesome”, “yipee”) produced in a neutral and in a positive-emotional prosody by the Amazon Alexa US-English default voice and by a human talker. They found that participants mirrored emotional productions by both human and TTS voices with parallel acoustic modifications: increasing duration, pitch, and pitch variation, mirroring the emotional adjustments.

However, independent analysis of each acoustic property can miss other patterns present in the data. For example, examination of just speech rate might not detect if there are patterns of convergence or divergence in intonation contour, or if the combination

of features leads to mirroring. This is an important limitation to address as degree of alignment does not occur uniformly across phonetic features, or even additional levels of the grammar (e.g., semantic, syntactic alignment) [45].

1.1.3 AXB Similarity Approaches to Test Similarity. To address the limitation of individual acoustic features in tracking mirroring, some groups have used external (human) raters to assess similarity using an AXB paradigm [46, 47, 56]. AXB asks listeners to determine whether “A” (a speaker’s pre-exposure production) or “B” (a speaker’s post-exposure production) sound more similar to “X” (their interlocutor’s production). If there is mirroring, then the likelihood of selecting “B” (post-exposure sounds more similar) should increase. For example, prior work has shown higher perceived mirroring using AXB for people interacting with other human addressees, compared to voice assistants [14, 21, 57]. Using the AXB paradigm, [21] found that speakers mirror emotionally expressive productions of words more than those words in a neutral tone produced by an Amazon’s Alexa TTS voice (but note that no human comparison was provided). [14] found differences in perceived mirroring toward human and Apple’s Siri TTS voices, with greater mirroring toward male than female voices overall, and more for the human than TTS voices. In another study, [57] found greater mirroring toward a male human voice than for an AWS Polly TTS voice using AXB. While the holistic approach allows for an overall comparison— one agnostic to a particular acoustic feature or cluster of features — one of its limitations is that it relies on the human listener. While in some cases listeners assess an entire set of productions in an experiment [16, 21], this is not possible for larger studies. Additionally, working memory constraints — in maintaining pronunciation patterns from “A”, “X”, and “B” — mean that the comparisons might be focused on one type of feature (e.g., overall duration or intonation contour) and might miss other fine-grained features.

1.1.4 Embedding Distance Approaches to Test Similarity. Recent advances in deep learning have led to novel approaches to measure linguistic similarity in word meaning [27, 41] as well as for audio recordings [2, 4, 5]. Deep learning based audio models transform audio signals into latent representations (often referred to as embeddings) that capture multidimensional information of the signal. Unlike feature-based approaches, these embeddings reflect the overall acoustic similarity between samples, creating more of a parallel with human AXB perceptual ratings experiments [46, 47]. Distance in latence space has been commonly applied to assess speakers’ pronunciations in other domains. For example, [2] used audio embeddings (wav2vec 2.0) to assess differences in speech style and pronunciation from raw audio in latent space; they found that certain hidden features in speech processing are specialized to differentiate phonetic properties. [5] assessed pronunciation by non-native English speakers (99 first languages, including Spanish, French, Arabic, Dutch) with native English speaker counterparts in the Speech Accent Archive using different models (wav2vec 2.0). They found a strong correlation between human listener perceptions and embedding distance of the wav2vec 2.0 English model, concluding that the acoustic embeddings from the middle layers of the transformer models retained features that closely reflected human perception of pronunciation differences. [28] compared audio

similarity scores by humans and by automatic speech recognition (ASR) systems and found that the ASR systems' scores were consistent with the listener judgments; this study used the DyViS database, which is comprised of 100 male speakers aged 18–25 years with a Standard Southern British English, with approximately 1.25 hours of speech per speaker. They used this audio with VOCALISE, a speaker recognition system, to extract auto-phonetic features from each speech sample, model the intra-speaker and inter-speaker variability, and score the likelihood of two vectors coming from the same or different speakers.

1.2 Current Study

The current study probes emotional mirroring using two methods: a human similarity perception using AXB (Experiment 1; Section 2) and distance in word embeddings (Experiment 2; Section 3). To draw comparisons across AXB and embedding methods, we selected data from a prior study [19], an experiment with clear acoustic adjustments that occur in one direction: increasing duration, pitch, and pitch variation following exposure to an interjection produced with positive-emotional prosody (e.g., “great!”) compared to the same interjection produced with more neutral prosody by both human and TTS talkers.

2 PERCEPTUAL SIMILARITY EXPERIMENT (HUMAN LISTENER)

2.1 Stimuli

Speaker recordings come from [19], consisting of $n=36$ native English speakers from the United States (mean age=20.86 years old, $sd=2.47$; female=19, male=17). While in the lab, speakers produced baseline productions of $n=10$ interjections (e.g., “great”, “awesome”). Then, they shadowed (i.e., repeated after) a human voice and the US-English Alexa (female) TTS voice producing the same words in a neutral and an emotionally expressive prosody (randomly presented).

2.2 Participants

A total of 109 participants (mean age=19.0 years old, $sd=2.62$, female=87, male=21, queer=1) completed the perceptual similarity ratings (AXB) study. Raters were native English speakers recruited from the UC Davis Psychology subjects' pool, and received course credit for their participation. None of the participants reported having a known hearing difficulty. The study was approved by the UC Davis Institutional Review Board (IRB) and participants completed informed consent before participating.

2.3 Stimuli

Stimuli were created by concatenating three sound files (amplitude normalized to 60 dB) in in Praat [8] with 400 milliseconds of silence between them:

- (A) the speaker's pre-exposure production of a word;
- (X) the interlocutor's production of the word (e.g., ‘awesome’ Alexa TTS, neutral emotion);
- (B) the speaker's post-exposure production of that word after shadowing the interlocutor in that condition.

Order of A (pre-exposure) and B (post-exposure) was counterbalanced across AXB sound files for each speaker.

2.4 Procedure

The experiment was conducted online, via a Qualtrics survey. On each trial, participants heard an AXB sound file and were asked to select which of the speaker's productions sounded more similar to the interlocutor's (“X”) production by clicking the corresponding choice on the screen (‘first’ or ‘third’). For example, if a participant heard (A) (pre-exposure) – 400 ms – (X) – 400 ms – (B) (post-exposure) and selected, ‘third’, this would indicate perceived mirroring. Raters had to make a selection before they could continue to the next trial. Raters were presented with all the items from each speaker, across the two interlocutor conditions (Alexa, human) and two prosody conditions (neutral, emotionally expressive). Each speaker's productions were assessed by a minimum of four raters [47, 56]. Twelve experiment versions were created, containing the full set of items from three speakers. Raters were randomly assigned to one of the versions. In total, listeners completed 120 trials (3 speakers * 10 words * 2 interlocutors * 2 prosody conditions) in random order.

2.5 Analysis and Results

Responses were coded for whether the post-exposure production was selected as more similar to the interlocutor's production (=1) or not (=0). We analyzed these responses with a mixed-effects logistic regression (lme4 R package; [6]). Fixed effects included Prosody (neutral, emotionally expressive), Interlocutor (Alexa, human), and their interaction. Random effects included by-Rater, by-Speaker, and by-Word random intercepts, as well as by-Speaker random slopes for Prosody and Interlocutor (and their interaction). Contrasts were sum coded.

Mean similarity responses are plotted in Figure 1.A. and 1.B. The full model output is in Appendix A. First, there was a significant intercept, such that listeners tended to select the post-exposure production [Coef = 0.43, $z = 4.75$, $p < 0.001$] at greater-than-chance likelihood (probability: 0.61). There was an effect of Prosody: as seen in Figure 1.A., raters were more likely to select the post-exposure production as more similar to the interlocutor in the expressive condition [Coef = 0.25, $z = 4.55$, $p < 0.001$]. Additionally, there was an effect of Interlocutor Type: as seen in Figure 1.B., participants mirrored the Alexa TTS voice less than human voice overall [Coef = -0.05, $z = -1.97$, $p < 0.05$]. No other effects or interactions were observed.

2.6 Interim Discussion

Experiment 1 showed an effect of emotional mirroring: speakers mirror emotionally expressive prosody more than neutral productions of words. This is consistent with an acoustic assessment of mirroring in [19] that found that shadowers increased their pitch, pitch variation, and duration following exposure of positive emotionally expressive words. Contrary to socially-mediated accounts, we see equivalent emotional mirroring toward human and TTS productions, both for individual acoustic features [19] and holistically by external raters (current study). Furthermore, independent of the emotional mirroring effects, participants mirror the Alexa

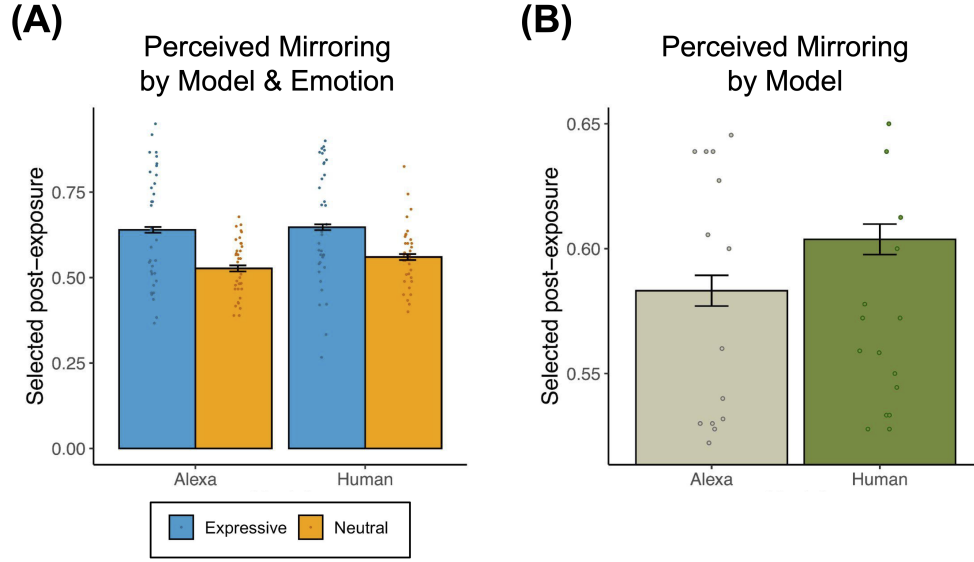


Figure 1: AXB Perceptual Similarity Ratings. Mean proportion of post-exposure token selection by (A). Interlocutor (Alexa, Human) and Prosody Condition (Expressive, Neutral). (B). By Interlocutor (Alexa, Human) (i.e., model talker) only. Error bars indicate standard errors of the mean. Individual points indicate speaker means for each condition.

TTS voice less than the human voice. This general finding parallels observations in related work, such as less mirroring of TTS voices than human voices [14, 31, 57], suggesting that, even with introductions of ‘emotion’, TTS voices are socially distinct from human voices.

3 EMBEDDING DISTANCE EXPERIMENT (WAV2VEC 2.0)

Experiment 2 assesses emotional mirroring using distance in latent space quantified by the wav2vec 2.0 model. Wav2vec 2.0 is a transformer-based model developed by Facebook AI Research that is primarily used to convert raw audio waveforms into embeddings, or vector representations of audio features. The wav2vec 2.0 model [2] was pre-trained on the LibriSpeech dataset, which consists of 960 hours of read speech in English, and fine-tuned for phoneme recognition on the TIMIT dataset, consisting of 630 speakers producing ten sentences, ranging across eight dialects of United States English. Wav2vec 2.0 has been previously used to assess similarity in pronunciation [2, 5], and can provide a holistic distance in latent space, serving as an apt comparison to human AXB similarity ratings methods in assessing emotional mirroring.

3.1 Stimuli

The identical stimuli were used from Experiment 1 (AXB Similarity study, described in Section 2.1).

3.2 Analysis

We used wav2vec 2.0 (English) to quantify mirroring in speakers’ productions using its feature extractor. For each production (pre-exposure, post-exposure, and interlocutor), we calculated the

word embedding in latent space based on 768 vectors. At each of the 12 layers, we calculated the cosine difference between the pre-exposure-to-interlocutor and post-exposure-to-interlocutor. We assigned a binary value of 1 to each target word if the cosine distance between the baseline-to-interlocutor was larger than the post-exposure-to-interlocutor, indicating perceived mirroring (=0 if not).

We modeled the likelihood of mirroring (=1) per embedding layer within the wav2vec 2.0 model with a mixed effects logistic regression model with the lme4 R package [6]. Fixed effects included Prosody (2 levels: neutral, emotionally expressive; sum coded), Interlocutor (2 levels: Alexa, human; sum coded), and layer (0-12; ordered factor, reference = Layer 0), and all possible interactions. The random effects included by-Speaker and by-Word random intercepts, as well as by-Speaker random slopes for Prosody and Interlocutor (and their interaction), as well as by-Speaker random slopes for Layer. In cases of singularity or convergence errors, we simplified the random effects structure following [3, 15].

3.3 Results

The model output is provided in Appendix B, including the retained model structure. First, we found an effect of Prosody: there is greater mirroring of emotionally expressive speech [Coef = 0.11, $z = 2.65$, $p < 0.01$], as seen in 2.A. We found no effects of Interlocutor or any interactions between Interlocutor and Prosody; that is, there was no difference in mirroring toward a human and TTS voice. We also found effects of Layer. As seen in Figure 2.B., results show that for individual layers, mirroring is relatively stronger during the early layers (1–5). For each individual layer, we found that the early layers had a positive fixed effect, the middle layer (6–7) effects were around zero, and the later layers (8–12) were negative. There

were no interactions between Layer and Prosody. However, we did observe one interaction between Layer and Interlocutor only for the first layer [Coef = 0.11, $z = 2.16$, $p < 0.05$], with stronger mirroring towards the TTS voice. Finally, there were no three-way interactions between Prosody, Interlocutor, and Layer.

3.4 Interim Discussion

In Experiment 2, we found that wav2vec 2.0 embeddings captured speakers' mirroring toward emotional prosody, paralleling Experiment 1, as well as a feature-based approach [19]. We also observed that in Layer 1, there was greater mirroring toward the Alexa TTS voice. Finally, we observed that overall mirroring is best detected in the earlier layers (Layers 1–5), paralleling [5] who found that middle layers best captured more native-like pronunciation by a group of non-native English speakers. These results correlate with the audio similarity score assessment by humans and ASR systems conducted by [28], where both confirm that the models agreed with human listeners. Furthermore, our findings are also consistent with the assessment between human listener perception and acoustic embeddings from transformer models on pronunciation differences in [5] that found that the extracted features from the hidden middle layers of the transformer models were optimal in distinguishing the phonetic variations in audio. Our results indicate that the earlier layers – rather than the middle layers – were the distinct layers that picked up on mirroring in the acoustic embeddings.

4 GENERAL DISCUSSION

This study provides new insight and methodological innovation in assessing people's vocal mirroring in interactions with other humans compared to technology. Speakers (all native English speakers from the United States) completed a controlled experiment [19] where they produced target words as 'baseline' prior to any interaction and 'shadowed' another talker producing them, repeating after they heard the word spoken in an expressive or neutral manner (e.g., "Awesome!" vs. "Awesome."). We used two methods to compare mirroring: by external human raters (Experiment 1) and by distance in embedded space (Experiment 2).

First, we see consistent evidence across both experiments that speakers mirror other talkers' emotional prosody. This effect was observed for both human raters (Experiment 1) and in embedded distance (Experiment 2). These approaches provide additional insight to individual acoustic feature-based approaches used previously (e.g., greater duration, pitch, and pitch range in [19]). That we observe parallel findings for global similarity for human raters and in latent space validates wav2vec 2.0 as a method to assess emotional mirroring. This extends related work showing that wav2vec 2.0 can be implemented to assess native-like pronunciation of words (e.g., [5]), suggesting that it can detect aspects of emotional prosody as well.

Second, in both AXB and embedding distance measures, we do not observe differences in the degree of emotional mirroring based on whether the talker was shadowing a word produced by a human or a TTS voice. This finding is consistent with matched motor accounts of emotional mirroring [22, 23, 48], which propose that people subconsciously mirror emotion regardless of its source. Additionally, that we observe parallel patterns in emotional mirroring

toward the human and TTS voices is consistent with theories of technology equivalence [37, 42, 43], which propose that social behaviors from human-human interaction are automatically applied to human-computer interaction given a cue of humanity. Broadly, this finding suggests that people's emotional mirroring is robust, both in response to a human as well as to a TTS voice producing it, and could have implications in the effectiveness of synthesized emotion (e.g., [13]).

Third, we observe slight differences in similarity measures by interlocutor across in Experiments 1 and 2. The AXB similarity analysis showed that speakers mirror the Alexa TTS voice less overall than the human voice. This finding is consistent with prior work reporting greater vocal mirroring of other human voices than TTS voices [14, 31, 57]. At the same time, Experiment 2 revealed greater perceived mirroring toward the Alexa voice, but only for the first layer of the wav2vec 2.0 embedding. Here, one possibility is that the wav2vec 2.0 model, trained on the Librispeech dataset of human voices [2], does not have as robust of representation for TTS voices and therefore does not capture mirroring as well as human listeners. At the same time, that wav2vec 2.0 does perceive a small increase in mirroring toward the TTS voice could suggest that there is a dimension that human listeners do not pick up on, which extends work showing largely similar patterns for humans and neural nets (e.g., speaker classification in [28]).

4.1 Limitations and Future Directions

This study has several limitations, which can serve as avenues for future research. First, the stimuli all consisted of positive-emotional interjections. Future work varying prosody across positive, negative, and neutral contexts can probe the extent emotional mirroring is equivalent across differing emotional expressiveness. Due to availability of emotionally expressive productions by the Alexa US-English female TTS voice during the time of the study, we constrained the interlocutors to only female talkers. The role of paralinguistic features of TTS voices, however, has shown to play a role in mirroring behaviors. Future work examining emotion produced by voices varying in apparent gender, age, and race/ethnicity can provide a fuller picture of this effect, as well as probe socially-mediated accounts of emotional mirroring [25, 32, 33]. Furthermore, the speakers in the present study were college-age adults in the United States; future research examining other participant groups can reveal the extent these findings might differ cross-culturally and cross-linguistically.

Additionally, the type of interaction – as a single word shadowing – might have led to more similar emotional mirroring patterns for the human and TTS voices. For example, in an interaction with less focus on the pronunciation (e.g., chatbot interactions in [12, 18]), speakers might have less monitoring of their productions. Moreover, this study compared mirroring toward emotionally expressive speech. The extent to which individual acoustic features, AXB similarity ratings, and word embeddings capture similar patterns for other types of mirroring (e.g., pronunciation) toward human and TTS voices is a ripe avenue for future study. Additionally, there is some work showing that embeddings can capture native speaker-like pronunciation patterns, particularly for second language (L2)

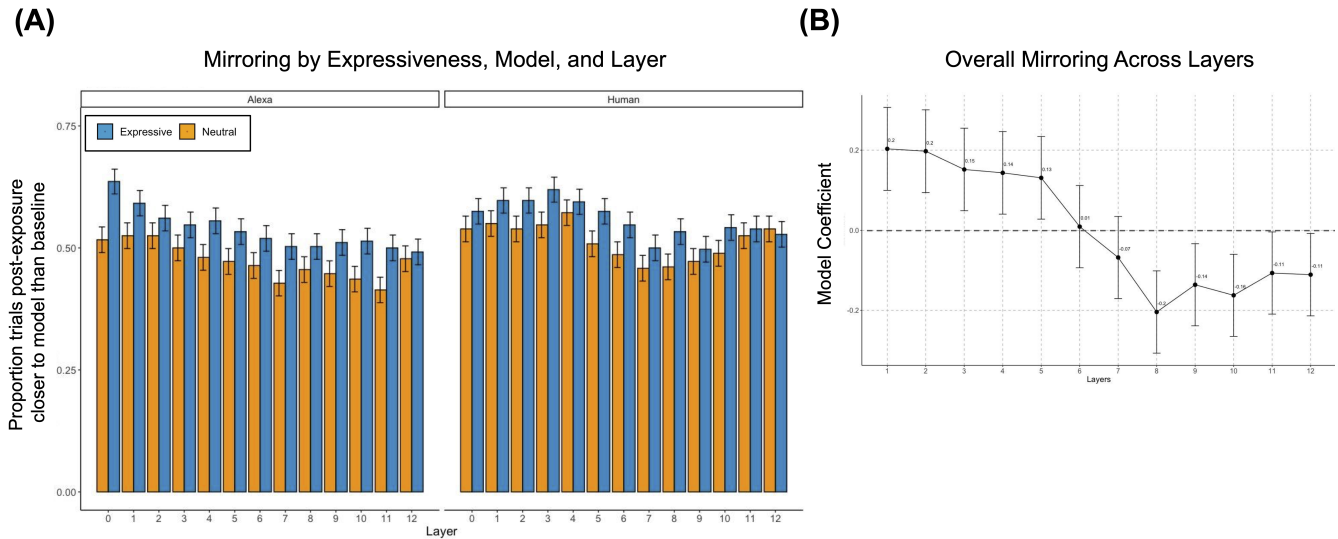


Figure 2: (A). Mean proportion of trials where the post-exposure production is closer to the model (i.e., interlocutor), relative to embedding distance from the baseline production to the model. Interlocutor (Alexa, Human) is faceted and Expressiveness Condition (Emotionally Expressive, Neutral) is shown in the bars (dark blue = Expressive, orange = Neutral). Error bars indicate standard errors of the mean. **(B).** Beta coefficients for each wav2vec 2.0 layer, relative to Layer 0 (collapsed across Interlocutor and Expressiveness Conditions). Error bars indicate standard errors.

speakers [5], suggesting embeddings might be fruitful for assessing phonetic mirroring of TTS voices.

Finally, we observed some differences in the embedding layers in capturing mirroring, with earlier layers better capturing mirroring than middle or late layers. Ultimately, it is an open question as to which specific features each layer of the wav2vec 2.0 model focus on, as well as which features both human perception and speech recognition systems discern.

5 CONCLUSION

Overall, this study contributes to our understanding of interpersonal dynamics in speech behavior, and provides evidence for an unmediated, matched motor account [22, 23, 48] shaping emotional mirroring of both human and device interlocutors. We additionally provide methodological advancements in assessing emotional mirroring in speech using distance in latent space (wav2vec 2.0). Examining both objective measures (embeddings) in comparison to human perceptual ratings (AXB similarity) can provide a more complete picture of the dynamics of mirroring.

REFERENCES

- [1] Pablo Arias, Pascal Belin, and Jean-Julien Aucouturier. 2018. Auditory smiles trigger unconscious facial imitation. *Current Biology* 28, 14 (2018), R782–R783.
- [2] Alexei Baevski, Yuhao Zhou, Abdelrahman Mohamed, and Michael Auli. 2020. wav2vec 2.0: A framework for self-supervised learning of speech representations. *Advances in neural information processing systems* 33 (2020), 12449–12460.
- [3] Dale J. Barr, Roger Levy, Christoph Scheepers, and Harry J. Tily. 2013. Random effects structure for confirmatory hypothesis testing: Keep it maximal. *Journal of memory and language* 68, 3 (2013), 255–278. <https://doi.org/10.1016/j.jml.2012.11.001>
- [4] Martijn Bartelds, Wietse de Vries, Caitlin Richter, Mark Liberman, and Martijn Wieling. 2021. Measuring foreign accent strength using an acoustic distance measure. In *12th International Seminar on Speech Production*. Haskins Press, 17–20.
- [5] Martijn Bartelds, Wietse de Vries, Faraz Sanal, Caitlin Richter, Mark Liberman, and Martijn Wieling. 2022. Neural representations for modeling variation in speech. *Journal of Phonetics* 92 (2022), 101137.
- [6] Douglas Bates, Martin Mächler, Ben Bolker, and Steve Walker. 2015. Fitting Linear Mixed-Effects Models Using lme4. *Journal of Statistical Software* 67, 1 (Oct. 2015), 1–48. <https://doi.org/10.18637/jss.v067.i01>
- [7] Marouane Birjali, Mohammed Kasri, and Abderrahim Beni-Hssane. 2021. A comprehensive survey on sentiment analysis: Approaches, challenges and trends. *Knowledge-Based Systems* 226 (2021), 107134.
- [8] Paul Boersma and David Weenink. 2021. Praat: doing phonetics by computer [computer program](2011). *Version 5*, 3 (2021), 74.
- [9] Holly P. Branigan, Martin J. Pickering, Jamie Pearson, Janet F. McLean, and Ash Brown. 2011. The role of beliefs in lexical alignment: Evidence from dialogs with humans and computers. *Cognition* 121, 1 (2011), 41–57. <https://doi.org/10.1016/j.cognition.2011.05.011>
- [10] Holly P. Branigan, Martin J. Pickering, Jamie Pearson, Janet F. McLean, and Clifford Nass. 2003. Syntactic alignment between computers and people: The role of belief about mental states. In *Proceedings of the 25th annual conference of the cognitive science society*. Lawrence Erlbaum Associates, 186–191. <http://csjarchive.cogsci.rpi.edu/proceedings/2003/pdfs/55.pdf>
- [11] Jessy Ceha and Edith Law. 2022. Expressive Auditory Gestures in a Voice-Based Pedagogical Agent. In *Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems*. 1–13.
- [12] Chun-Yen Chen, Dian Yu, Weiming Wen, Yi Mang Yang, Jiaping Zhang, Mingyang Zhou, Kevin Jesse, Austin Chau, Antara Bhowmick, and Shreenath Iyer. 2018. Gunrock: Building A Human-Like Social Bot By Leveraging Large Scale Real User Data. *2nd Proceedings of Alexa Prize* (2018). <http://dex-microsites-prod.s3.amazonaws.com/alexaprize/2018/papers/Gunrock.pdf>
- [13] Michelle Cohn, Chun-Yen Chen, and Zhou Yu. 2019. A Large-Scale User Study of an Alexa Prize Chatbot: Effect of TTS Dynamism on Perceived Quality of Social Dialog. In *Proceedings of the 20th Annual SIGDial Meeting on Discourse and Dialogue*. Stockholm, Sweden, 293–306. <https://www.sigdial.org/files/workshops/conference20/proceedings/cdrom/pdf/W19-5935.pdf>
- [14] Michelle Cohn, Bruno Ferenc Segedin, and Georgia Zellou. 2019. Imitating Siri: Socially-mediated alignment to device and human voices. In *Proceedings of International Congress of Phonetic Sciences*. Melbourne, Australia, 1813–1817. https://icphs2019.org/icphs2019-fullpapers/pdf/full-paper_202.pdf
- [15] Michelle Cohn, Bruno Ferenc Segedin, and Georgia Zellou. 2022. Acoustic-phonetic properties of Siri- and human-directed speech. *Journal of Phonetics* 90 (Jan. 2022), 101123. <https://doi.org/10.1016/j.wocn.2021.101123>

- [16] Michelle Cohn, Patrik Jonell, Taylor Kim, Jonas Beskow, and Georgia Zellou. 2020. Embodiment and gender interact in alignment to TTS voices. In *Proceedings of the Cognitive Science Society*. Toronto, Canada, 220–226. <https://cogsci.mindmodeling.org/2020/papers/0044/0044.pdf>
- [17] Michelle Cohn, Ashley Keaton, Jonas Beskow, and Georgia Zellou. 2023. Vocal accommodation to technology: the role of physical form. *Language Sciences* 99 (2023), 101567.
- [18] Michelle Cohn, Kai-Hui Liang, Melina Sarian, Georgia Zellou, and Zhou Yu. 2021. Speech Rate Adjustments in Conversations With an Amazon Alexa Socialbot. *Frontiers in Communication* 6 (2021), 1–8. <https://doi.org/10.3389/fcomm.2021.671429>
- [19] Michelle Cohn, Kristin Predeck, Melina Sarian, and Georgia Zellou. 2021. Prosodic alignment toward emotionally expressive speech: Comparing human and Alexa model talkers. *Speech Communication* 135 (Dec. 2021), 66–75. <https://doi.org/10.1016/j.specom.2021.10.003>
- [20] Michelle Cohn, Eran Raveh, Kristin Predeck, Iona Gessinger, Bernd Möbius, and Georgia Zellou. 2020. Differences in Gradient Emotion Perception: Human vs. Alexa Voices. *Proceedings of Interspeech 2020, 21st Conference of the International Speech Communication Association* (2020), 1818–1822. <https://doi.org/10.21437/Interspeech.2020-1938>
- [21] Michelle Cohn and Georgia Zellou. 2019. Expressiveness influences human vocal alignment toward voice-AI. In *Proceedings of Interspeech 2019, 20th Conference of the International Speech Communication Association*. Graz, Austria, 41–45. <https://doi.org/10.21437/Interspeech.2019-1368>
- [22] Frans BM De Waal. 2007. The ‘Russian doll’ model of empathy and imitation. *On being moved: From mirror neurons to empathy* (2007), 35–48.
- [23] Jean Decety and Philip L. Jackson. 2006. A social-neuroscience perspective on empathy. *Current directions in psychological science* 15, 2 (2006), 54–58.
- [24] Andrew Fandrianto and Maxine Eskenazi. 2012. Prosodic entrainment in an information-driven dialog system. In *Thirteenth Annual Conference of the International Speech Communication Association*.
- [25] Agneta H. Fischer, Lisanne S. Pauw, and Antony S. R. Manstead. 2019. *Emotion Recognition as a Social Act: The Role of the Expresser-Observer Relationship in Recognizing Emotions*. Springer International Publishing, Cham, 7–24. https://doi.org/10.1007/978-3-030-32968-6_2
- [26] Cindy Gallois and Howard Giles. 2015. Communication accommodation theory. *The international encyclopedia of language and social interaction* (2015), 1–18. <https://doi.org/10.1002/9781118611463.wbielsi066>
- [27] Nikhil Garg, Londa Schiebinger, Dan Jurafsky, and James Zou. 2018. Word embeddings quantify 100 years of gender and ethnic stereotypes. *Proceedings of the National Academy of Sciences* 115, 16 (2018), E3635–E3644.
- [28] Linda Gerlach, Kirsty McDougall, Finnian Kelly, Anil Alexander, and Francis Nolan. 2020. Exploring the relationship between voice similarity estimates by listeners and by an automatic speaker recognition system incorporating phonetic features. *Speech Communication* 124 (2020), 85–95.
- [29] Iona Gessinger, Michelle Cohn, Georgia Zellou, and Bernd Möbius. 2022. Cross-Cultural Comparison of Gradient Emotion Perception: Human vs. Alexa TTS Voices. *Proceedings of Interspeech 2022, 23rd Conference of the International Speech Communication Association* (2022), 4970–4974.
- [30] Iona Gessinger, Eran Raveh, Sébastien Le Maguer, Bernd Möbius, and Ingmar Steiner. 2017. Shadowing Synthesized Speech-Segmental Analysis of Phonetic Convergence. In *Proceedings of Interspeech 2017, 18th Conference of the International Speech Communication Association*. 3797–3801. <https://doi.org/10.21437/Interspeech.2017-1433>
- [31] Iona Gessinger, Eran Raveh, Ingmar Steiner, and Bernd Möbius. 2021. Phonetic accommodation to natural and synthetic voices: Behavior of groups and individuals in speech shadowing. *Speech Communication* 127 (2021), 43–63. <https://doi.org/10.1016/j.specom.2020.12.004>
- [32] Ursula Hess and Agneta Fischer. 2013. Emotional mimicry as social regulation. *Personality and social psychology review* 17, 2 (2013), 142–157.
- [33] Ursula Hess and Agneta Fischer. 2014. Emotional mimicry: Why and when we mimic emotions. *Social and Personality Psychology Compass* 8, 2 (2014), 45–57.
- [34] Jiaxiong Hu, Yun Huang, Xiaozhu Hu, and Yingqing Xu. 2021. Enhancing the perceived emotional intelligence of conversational agents through acoustic cues. In *Extended Abstracts of the 2021 CHI Conference on Human Factors in Computing Systems*. 1–7.
- [35] Jacqueline M. Kory-Westlund and Cynthia Breazeal. 2019. Exploring the effects of a social robot’s speech entrainment and backstory on young children’s emotion, rapport, relationship, and learning. *Frontiers in Robotics and AI* 6 (2019), 54.
- [36] Jessica L. Lakin, Valerie E. Jefferis, Clara Michelle Cheng, and Tanya L. Chartrand. 2003. The chameleon effect as social glue: Evidence for the evolutionary significance of nonconscious mimicry. *Journal of nonverbal behavior* 27, 3 (2003), 145–162.
- [37] Kwan Min Lee. 2008. *Media Equation Theory*. Vol. 1. John Wiley & Sons, Ltd, Malden, MA, USA, 1–4. <https://doi.org/10.1002/97811405186407.wbiecm035>
- [38] Rivka Levitan, Stefan Benus, Ramiro H. Gálvez, Agustín Gravano, Florencia Savoretti, Marian Trnka, Andreas Weise, and Julia Hirschberg. 2016. Implementing Acoustic-Prosodic Entrainment in a Conversational Avatar. In *Interspeech*, Vol. 16. San Francisco, CA, 1166–1170.
- [39] Nichola Lubold and Heather Pon-Barry. 2014. Acoustic-prosodic entrainment and rapport in collaborative learning dialogues. In *Proceedings of the 2014 ACM workshop on Multimodal Learning Analytics Workshop and Grand Challenge*. Istanbul, Turkey, 5–12. <https://doi.org/10.1145/2666633.2666635>
- [40] Nichola Lubold, Erin Walker, Heather Pon-Barry, and Amy Ogan. 2018. Automated pitch convergence improves learning in a social, teachable robot for middle school mathematics. In *Artificial Intelligence in Education: 19th International Conference, AIED 2018, London, UK, June 27–30, 2018, Proceedings, Part I* 19. Springer, 282–296.
- [41] Julia Mendelsohn, Yulia Tsvetkov, and Dan Jurafsky. 2020. A Framework for the Computational Linguistic Analysis of Dehumanization. *Frontiers in Artificial Intelligence* 3 (2020), 55. <https://doi.org/10.3389/frai.2020.00055>
- [42] Clifford Nass, Youngme Moon, John Morikes, Eun-Young Kim, and B. J. Fogg. 1997. Computers are social actors: A review of current research. *Human values and the design of computer technology* 72 (1997), 137–162.
- [43] Clifford Nass, Jonathan Steuer, and Ellen R. Tauber. 1994. Computers are social actors. In *Proceedings of the SIGCHI conference on Human factors in computing systems*. ACM, Boston, Massachusetts, USA, 72–78. <https://doi.org/10.1145/259963.260288>
- [44] Costanza Navarretta. 2016. Mirroring facial expressions and emotions in dyadic conversations. In *Proceedings of the Tenth International Conference on Language Resources and Evaluation (LREC’16)*. 469–474.
- [45] Rachel Ostrand and Eleanor Chodoff. 2021. It’s alignment all the way down, but not all the way up: Speakers align on some features but not others within a dialogue. *Journal of Phonetics* 88 (Sept. 2021), 101074. <https://doi.org/10.1016/j.wocn.2021.101074>
- [46] Jennifer Pardo. 2013. Measuring phonetic convergence in speech production. *Frontiers in Psychology* 4 (2013), 559. <https://doi.org/10.3389/fpsyg.2013.00559>
- [47] Jennifer S. Pardo, Adelya Urmanche, Sherilyn Wilman, and Jaclyn Wiener. 2017. Phonetic convergence across multiple measures and model talkers. *Attention, Perception, & Psychophysics* 79, 2 (Feb. 2017), 637–659. <https://doi.org/10.3758/s13414-016-1226-0>
- [48] Stephanie D. Preston. 2007. A perception-action model for empathy. *Empathy in mental illness* (2007), 428–447.
- [49] Eran Raveh, Ingo Siegert, Ingmar Steiner, Iona Gessinger, and Bernd Möbius. 2019. Three’s a Crowd? Effects of a Second Human on Vocal Accommodation with a Voice Assistant. *Proc. Interspeech 2019* (2019), 4005–4009. <https://doi.org/10.21437/Interspeech.2019-1825>
- [50] Najmeh Sadoughi, André Pereira, Rishub Jain, Iolanda Leite, and Jill Fain Lehman. 2017. Creating prosodic synchrony for a robot co-player in a speech-controlled game for children. In *Proceedings of the 2017 ACM/IEEE International Conference on Human-Robot Interaction*. 91–99.
- [51] Nastaran Saffaryazdi, Yenuska Goonesekera, Nafiseh Saffaryazdi, Nebiyou Daniel Hailemariam, Ebasa Girma Temesgen, Suranga Nanayakkara, Elizabeth Broadbent, and Mark Billinghurst. 2022. Emotion recognition in conversations using brain and physiological signals. In *27th International Conference on Intelligent User Interfaces*. 229–242.
- [52] Nastaran Saffaryazdi, Syed Talal Wasim, Kuldeep Dileep, Alireza Farrokhi Nia, Suranga Nanayakkara, Elizabeth Broadbent, and Mark Billinghurst. 2022. Using facial micro-expressions in combination with EEG and physiological signals for emotion recognition. *Frontiers in Psychology* 13 (2022), 864047.
- [53] Cathryn Snyder, Michelle Cohn, and Georgia Zellou. 2019. Individual variation in cognitive processing style predicts differences in phonetic imitation of device and human voices. In *Proceedings of Interspeech 2019, 20th Conference of the International Speech Communication Association*. Graz, Austria, 116–120. <https://doi.org/10.21437/Interspeech.2019-2669>
- [54] Jesse Thomason, Huy V. Nguyen, and Diane Litman. 2013. Prosodic entrainment and tutoring dialogue success. In *International conference on artificial intelligence in education*. Springer, 750–753.
- [55] Mayur Wankhade, Annavarapu Chandra Sekhara Rao, and Chaitanya Kulkarni. 2022. A survey on sentiment analysis methods, applications, and challenges. *Artificial Intelligence Review* 55, 7 (2022), 5731–5780.
- [56] Georgia Zellou, Michelle Cohn, and Bruno Ferenc Segedin. 2021. Age- and Gender-Related Differences in Speech Alignment Toward Humans and Voice-AI. *Frontiers in Communication* 5 (2021), 1–11. <https://doi.org/10.3389/fcomm.2020.600361>
- [57] Georgia Zellou, Michelle Cohn, and Tyler Kline. 2021. The influence of conversational role on phonetic alignment toward voice-AI and human interlocutors. *Language, Cognition and Neuroscience* 10 (May 2021), 1–15. <https://doi.org/10.1080/23273798.2021.1931372>

A EXPERIMENT 1 MODEL OUTPUT

Table 1: AXB Similarity Rating Model Output

Predictor	Estimate	Std. Error	z value	p value
(Intercept)	0.43	0.09	4.75	<0.001
Prosody(expressive)	0.25	0.05	4.55	<0.001
Interlocutor(Alexa)	-0.05	0.02	-1.97	<0.05
Prosody(expressive):Interlocutor(Alexa)	0.02	0.02	1.12	0.26
<i>Random effects</i>	<i>Variance</i>			
Speaker	0.23			
Speaker:Prosody(expressive)	0.09			
Speaker:Interlocutor(Alexa)	0.01			
Speaker:Prosody:Interlocutor	3.8e-03			
Rater	0.02			
Word	0.01			
<i>Num. observations = 12,840, Num. raters = 107, Num. speakers = 36, Num. words = 10</i>				

B EXPERIMENT 2 MODEL OUTPUT

Table 2: Embedding Distance Model Output

Predictor	Estimate	Std. Error	z value	p value
(Intercept)	0.09	0.05	1.62	0.11
Prosody(expressive)	0.11	0.04	2.65	<0.01
layer1	0.2	0.05	3.85	<0.001
layer2	0.2	0.05	3.75	<0.001
layer3	0.15	0.05	2.89	<0.01
layer4	0.14	0.05	2.73	<0.01
layer5	0.13	0.05	2.49	<0.05
layer6	0.01	0.05	0.17	0.86
layer7	-0.07	0.05	-1.3	0.19
layer8	-0.2	0.05	-3.9	<0.001
layer9	-0.14	0.05	-2.6	<0.01
layer10	-0.16	0.05	-3.1	<0.01
layer11	-0.11	0.05	-2.04	<0.05
layer12	-0.11	0.05	-2.11	<0.05
Interlocutor(Alexa)	-0.07	0.04	-1.85	0.06
Prosody(expressive):layer1	0.06	0.05	1.09	0.28
Prosody(expressive):layer2	0.01	0.05	0.2	0.84
Prosody(expressive):layer3	-0.01	0.05	-0.22	0.82
Prosody(expressive):layer4	0.02	0.05	0.29	0.77
Prosody(expressive):layer5	-0.01	0.05	-0.17	0.86
Prosody(expressive):layer6	0.02	0.05	0.42	0.67
Prosody(expressive):layer7	0.01	0.05	0.18	0.86
Prosody(expressive):layer8	0.01	0.05	0.18	0.86
Prosody(expressive):layer9	0.01	0.05	0.23	0.82
Prosody(expressive):layer10	-0.02	0.05	-0.39	0.7
Prosody(expressive):layer11	0.02	0.05	0.46	0.64
Prosody(expressive):layer12	-0.01	0.05	-0.15	0.88
Prosody(expressive):Interlocutor(Alexa)	0.02	0.05	0.45	0.65
layer1:Interlocutor(Alexa)	0.11	0.05	2.16	<0.05
layer2:Interlocutor(Alexa)	0.04	0.05	0.69	0.49
layer3:Interlocutor(Alexa)	0.01	0.05	0.27	0.78
layer4:Interlocutor(Alexa)	-0.06	0.05	-1.15	0.25
layer5:Interlocutor(Alexa)	-0.07	0.05	-1.35	0.18
layer6:Interlocutor(Alexa)	-0.02	0.05	-0.3	0.77
layer7:Interlocutor(Alexa)	0.01	0.05	0.26	0.8
layer8:Interlocutor(Alexa)	0.04	0.05	0.69	0.49
layer9:Interlocutor(Alexa)	0.03	0.05	0.53	0.6
layer10:Interlocutor(Alexa)	0.05	0.05	1.04	0.3
layer11:Interlocutor(Alexa)	-0.02	0.05	-0.37	0.71
layer12:Interlocutor(Alexa)	-0.09	0.05	-1.78	0.07
Prosody(expressive):layer1:Interlocutor(Alexa)	0.07	0.05	1.36	0.17
Prosody(expressive):layer2:Interlocutor(Alexa)	-1.3e-04	0.05	-2.0e-03	1
Prosody(expressive):layer3:Interlocutor(Alexa)	-0.05	0.05	-0.87	0.38
Prosody(expressive):layer4:Interlocutor(Alexa)	-0.05	0.05	-0.95	0.34
Prosody(expressive):layer5:Interlocutor(Alexa)	0.03	0.05	0.64	0.52
Prosody(expressive):layer6:Interlocutor(Alexa)	-0.03	0.05	-0.54	0.59
Prosody(expressive):layer7:Interlocutor(Alexa)	-0.03	0.05	-0.53	0.59
Prosody(expressive):layer8:Interlocutor(Alexa)	0.01	0.05	0.26	0.8
Prosody(expressive):layer9:Interlocutor(Alexa)	-0.05	0.05	-0.93	0.35
Prosody(expressive):layer10:Interlocutor(Alexa)	0.02	0.05	0.37	0.71
Prosody(expressive):layer11:Interlocutor(Alexa)	4.6e-03	0.05	0.09	0.93

Table 2 continued from previous page

Prosody(expressive):layer12:Interlocutor(Alexa)	0.05	0.05	1.04	0.3
<i>Random effects</i>	<i>Variance</i>			
Speaker	0.08			
Speaker:Prosody(expressive)	0.06			
Speaker:Interlocutor(Alexa)	0.04			
Speaker:Prosody:Interlocutor	0.08			
Word	4.0e-03			
<i>Num. observations = 18,720, Num. speakers = 36, Num. words = 10</i>				
